Conflict and Peace

Week 1: Introduction to the course and empirical analysis

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

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Roadmap for today

1. Introduction

1.1 Course logistics

1.2 Topics covered in course

1.3 Course grading, tasks, and timeline

1.4 Rules and Tips

2. Empirical Analysis

Course logistics

- Instructor: Seung-hun Lee
  - Office location: Room C06 in this building
  - Office hours: Thursdays 3:30-5:30PM (Sign up here)
  - From South Korea
  - Ph.D. in Economics from Columbia University (2024)
- TA: Su Mon Htet Naing
  - Contact: ekaterina.naing@gmail.com
- Class time and location
  - Time: Fridays 9AM - 12PM
  - Location: Room A09
- Coming prepared to the lecture
  - Lecture notes are uploaded a day in advance (No separate textbooks)
  - Keep up with the readings: more on that later

Course goal: Viewing politics from an economic perspective

- Focus on political economy of development related to conflict and peace
  - Conflicts: Violent events that disrupt economic and social stability - crime, political violence, civil wars, inter-state wars
  - Peace: Absence of conflict, or development of economic and institutional arrangements that prevents conflict

- Economic perspective: Making causal inferences between events/outcomes
  - Understand the theories behind incentives of stakeholders involved
  - Able to make educated, empirical analysis that identifies causal relations

- Goals and requirements
  - Be able to critically engage with scholarly work, conduct independent empirical analysis, and develop original research
  - Familiarity with statistical inference is necessary!
  - *Helpful: Graduate-level micro/macroeconomics*

Course topics (Refer to syllabus for full reading list)

- Course introduction and empirical analysis tools (Week 1)
- Cause of conflicts (Week 2-4)
  - *How do economic incentives drive individuals and states to conflict?*
  - *How do political and ethnic dynamics lead to conflict?*
  - *How and why do people engage in criminal activites?*
- Consequences of conflicts (Week 5-7)
  - *How do they affect individuals?*
  - *How do they shape economic development?*
  - *How do they alter the institutional development of the state?*

Course topics (Refer to syllabus for full reading list)

- Responses to conflicts (Week 9-10)
  - *How do individuals overcome collective action problems?*
  - *What drives people to migrate?*
  - *Hoow do people integrate to the new society?*
- State capacity (Week 11-13)
  - *Why do some states fail?*
  - *How do capable states build and maintain fiscal capacity?*
  - *How do capable states manage personnel capacity?*

Tasks and grading scale

- Weekly snap presentation (10% of the grades)
  - Two students will give a 15-minute presentation on two papers from the reading list marked with double asterisks
  - The presenters will be chosen at random on the day of the class **- So everyone must prepare the slides before the class**
  - Use **no more** than 8 slides, upload slides to eeclass after class

- Replication exercise (20% of the grades) - Due on Nov 21st
  - You are asked to use R or Stata to replicate Brodeur (2018), or Campante, Do (2014) - further details on these papers in syllabus
  - You will produce key results (in tables and figures) from one of those papers, and submit a **replication report** and **codes**
  - Report whether the data is accessible, and the key findings replicate.
  - More important: **Make your own extension -** alternative regression specifications, different independent variables, different falsification tests

Tasks and grading scale

- Virtual paper (50% of the grades)
  - Develop a viable paper plan that clearly motivates the research question, explains the background and data, empirical strategy, and preliminary findings
  - Does not need to be a complete paper **- A half-finished but original and exciting project is preferred** over a complete yet uninteresting one
  - Start with a two-page proposal (10% of the grades) - Due October 24
  - Then, 30-minute presentation on the full paper (10% of the grades) - December 5th
  - Based on comments, finish the draft (30% of the grades) - December 21st

- Final Exam (20% of the grades) - December 19th
  - Cumulative exam that will cover papers in single/double asterisks
  - Exact format to be determined soon

More on the virtual paper

- Format of the virtual paper: Mimics many scholarly articles
    1. Introduction: What is your research question? Why is it important from a scholarly or policy standpoint?
    2. Literature review: What is the gap in the literature that you relate your work to? How are you filling that gap?
    3. Background: What is the relevant information on the context that you are studying?
    4. Data: What data source are you using or planning to use? Why should it be that dataset?
    5. Empirical strategy: What statistical methods do you use? How do they allow for causal inference? What are the potential threats to that inference?
    6. Preliminary results: What are the statistical relationships that you have found so far?
    7. Mechanism and falsification tests: What factors may explain your main results? What are your plans to address potential factors that may nullify your findings?

Important deadlines

- October 24th: Presentation on 2-page proposal (submit draft before class)
- November 21st: Replication exercise (submit report and codes by email)
- December 5th: Virtual paper presentation
- (Week of) December 19th: Final Exam (exact time TBA)
- December 21st: Submit draft of the virtual paper by midnight

Some ground rules

- Usage of AI: Strictly for assistive purposes
  - Allowed for proofreading, coding support
  - Do not generate complete assignments using AI
  - All writings, especially virtual paper, must be original
  - To show that you are working on your own idea, regularly discuss your ideas with me through office hours
- Collaboration: You can collaborate, under these two conditions
  - Each student must write up their own assignments
  - Students must clearly identify their collaborators

Causal inference in social sciences: Quantifying causal relationships

- In this class (and beyond), we aim to clarify quantitative, causal relationship between different events and outcomes
  - *Does higher commodity prices reduce wars? (Caselli et al 2015; Dube, Vargas 2013)*
  - *Does extractive colonial institutional backdrop hurt current institutional development? (Acemoglu et al 2001; Michalopoulos, Papaiaonnou 2016)*
  - *Does negative shocks to economic productivity lead to revolutions (Brückner, Ciccone 2011)*
- Treatment effect measures the size of impact these events have on sample of interest
  - Identify an event that induces some shocks to individual behavior/societal outcome
  - Find a suitably comparable sample (i.e. the control group and treatment)
  - Compare outcomes for those affected by the event to those not affected
- To that end, we collect data from a suitably defined population and use various methods to estimate the effects of these events.

Correlation vs Causation

- Suppose you have random variables X and Y. You want to identify if X *causes* Y

- A **correlation** between X and Y is a statistical measure that describes how the two variables move together
  - It captures *any* type of statistical dependence that moves the two variables together: Causation, but also others too!
  - Not causal 1: Y can cause X (reverse causality)
  - Not causal 2: X and Y co-move because of Z that affects both (Omitted variable bias)

- A **causal** relationship: *Cleanly (exogenously)* changing variables X leads to changes in Y
  - Changes in X may be a combination of changes from X alone + other factors that co-move with X and affect Y (You do not want the latter)
  - Randomized experiments: One way of Isolating clean changes in X
  - Econometrics: Concisely explains the relationship between X and Y variables in a single equation (Even when experiment is not feasible)

Treatment effect in mathematics

- Mathematical notations
  - Y_i: observed outcome for individual i
  - $D_i = d$: Treatment assignment ($d = 1$ for treated, 0 for control group)
  - $Y_i(d)$: Outcome for treated individual if $d = 1$, control group if $d = 0$.

- Treatment effect: How much does outcome differ between treated and control group

$$TE = Y_i(1) - Y_i(0)$$

  - With large number of i, we want an average of TE (henceforth average treatment effect)

$$ATE = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)]$$

  - Fundamental problem: If i is treated, it cannot be in the control group
  - So either one of $Y_i(1)$ or $Y_i(0)$ is missing! $\rightarrow$ We do not exactly know the **counterfactual**
  - Econometrics provides ways to provide sound judgement on what the counterfactual looks like

Potential outcomes and randomized experiments

- Potential outcomes provides framework to match observations to treatment effect

$$\begin{aligned} Y_i &= Y_i(1)D_i + Y_i(0)(1 - D_i) \\ &= Y_i(0) + D_i(Y_i(1) - Y_i(0)) \end{aligned}$$

- In terms of conditional expectations,

$$\begin{aligned} E[Y_i|D_i] &= E[Y_i(0) + D_i(Y_i(1) - Y_i(0))|D_i] \\ &= E[Y_i(0)|D_i] + E[Y_i(1) - Y_i(0)|D_i]D_i \end{aligned}$$

- Note that
  - If $D_i = 1$, then $Y_i = Y_i(1)$ and $E[Y_i|D_i = 1] = E[Y_i(1)|D_i = 1]$
  - If $D_i = 0$, then $Y_i = Y_i(0)$ and $E[Y_i|D_i = 0] = E[Y_i(0)|D_i = 0]$

Potential outcomes and randomized experiments

- With randomized experiments, we assume that the counterfactual is identical **except** for treatment status
  - Mathematically, treatment assignment is independent of potential outcomes

$$E[Y_i(d)] = E[Y_i(d)|D_i = 1] = E[Y_i(d)|D_i = 0] \text{ for } d \in \{0, 1\}$$

  - This implies that

$$\begin{aligned} ATE &= E[Y_i(1)] - E[Y_i(0)] \\ &= E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] \\ &= E[Y_i|D_i = 1] - E[Y_i|D_i = 0] (\because \text{ previous slide}) \end{aligned}$$

    or that ATE is obtained by subtracting average value of Y_i for treatment group by that of control group (which we can do with the observed data we have)

Ordinary Least Square (OLS): Mathematical approach

- Write

$$Y_i = \alpha + \beta X_i + u_i$$

  where X_i is an independent variable and u_i is the error (determinants of Y_i not captured by X_i)

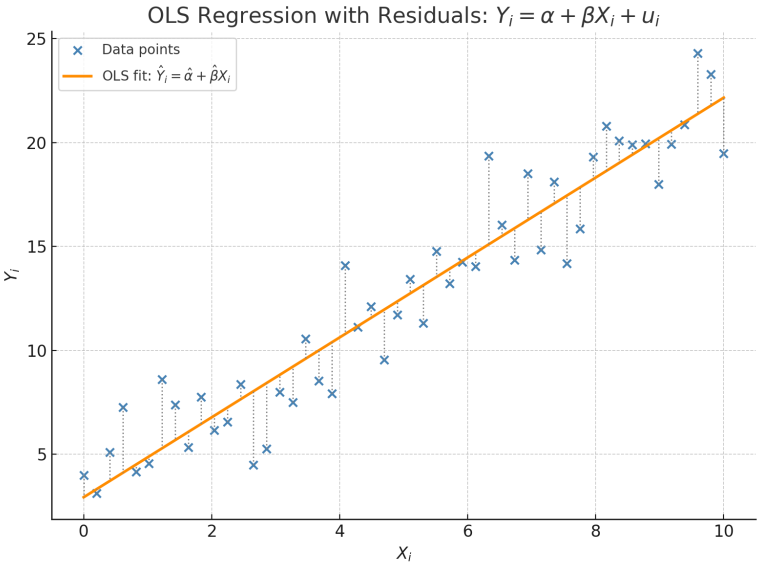
- Exogeneity: We assume that u_i is uncorrelated with X_i, $E[u_i|X_i] = 0$ (very important!)

- OLS estimates the parameter of interest β by minimizing the sum squared of errors, giving us the β that best matches the patterns in the observation

$$\hat{\beta}_{OLS} = \min_{\beta} \sum_{i=1}^{n} (Y_i - \alpha - \beta X_i)^2$$

  - We can do the same for α, which capture baseline average (average when all independent variables are zero)

Conceptualizing OLS



Ordinary Least Square (OLS): Potential outcomes approach

- Recall the potential outcome framework

$$
\begin{aligned}
Y_i &= Y_i(0) + D_i(Y_i(1) - Y_i(0)) \\
E[Y_i|D_i] &= E[Y_i(0)|D_i] + E[Y_i(1) - Y_i(0)|D_i]D_i
\end{aligned}
$$

- Combined with OLS regression

$$
\begin{aligned}
Y_i &= \alpha + \beta D_i + u_i \\
\implies E[Y_i|D_i] &= \alpha + \beta D_i + E[u_i|D_i] \\
&= \alpha + \beta D_i
\end{aligned}
$$

  - With random assignment, $ATE = E[Y_i(1) - Y_i(0)|D_i] = E[Y_i(1) - Y_i(0)] = \beta$
  - Thus, ATE can be obtained with OLS estimates of β, or $\hat{\beta}_{OLS}$

- You can add more variables, include X_i on top of D_i for better precision

Internal and external validity problems: Why OLS is not always usable

- **External validity** is concerned with the applicability to other contexts.

- **Internal validity** assesses whether the model accurately represents the sample
  - **Omitted variable bias:** Did the researcher left out factors that could affect Y and are correlated with X? $\rightarrow$ Multivariate regression, panel regression
  - **Wrong functional form:** Did the researcher incorporate X in a proper functional form? (quadratic, logs?) $\rightarrow$ add logs, or use logit/probit
  - **Measurement error:** If X does not have an accurate measure, OLS results in attenuation bias $\rightarrow$ Instrumental variable
  - **Sample selection bias:** If certain groups are more likely to not respond, then the data fails to estimate the true effects $\rightarrow$ Instrumental variable, Natural experiments
  - **Simultaneous causality bias:** There are cases where Y causes X too, also leading to the bias (consider supply and demand) $\rightarrow$ Instrumental variable

- Controlled experiment nearly impossible in political economy $\rightarrow$ Natural experiment that uses external events/shock as treatment

When exogeneity fails

- $E[u_i|D_i] = 0$ rarely holds: Treatment is not always random, or some individuals are more likely to be treated than others
  - This biases our results: Suppose that $E[u_i|D_i] = \phi D_i + v_i$ (u_i and D_i have correlation)

$$\begin{aligned} E[Y_i|D_i] &= \alpha + \beta D_i + E[u_i|D_i] \\ &= \alpha + (\beta + \phi) D_i + v_i \end{aligned}$$

With OLS on Y_i and D_i, $\hat{\beta}$ captures $\beta + \phi$ - Treatment effect and endogeneity bias

- When and why do biases happen?
  - Treated groups may be a good performer to begin with (and vice versa)
  - Differences in baseline characteristics besides treatment $\rightarrow$ inaccurate TE estimates
- Now what? Control for ϕ with individual/time indicators (Panel), exogenous proxy (IV), or natural experiments

Panel regression fixed effects approach

- We observe multiple individuals for multiple time periods ($i \in \{1, ..., N\}$, $t \in \{1, ..., T\}$)

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + ... + \beta_k X_{k,it} + u_{it}$$

- Problem: Unobserved individual characteristics and time trends could affect outcomes and cause omitted variable bias
  - Individual's unobserved ability/characteristics (e.g. culture)
  - A bad shock in year t that affects everyone (COVID years)

- Panel regression allows us to capture these using individual and time fixed effects

$$Y_{it} = \beta_1 X_{1,it} + ... + \beta_k X_{k,it} + \alpha_i + \lambda_t + u_{it}$$

  - α_i: Dummy for each individual $i \in \{1, ..., N\}$
  - δ_t: Dummy for each time $t \in \{1, ..., T\}$

Instrumental variables: Exogeneity amidst endogeneity

- OLS hinges on independent variable X_i being uncorrelated with u_i
  - This often breaks: omitted variable bias, measurement error, and reverse causality etc..
  - Parts of X_i that are correlated with u_i: What we want is part of X_i independent of u_i.
- Solution is to find another variable Z_i that are
  - (1) Correlated with X but (2) not with u: This gets us part of X_i independent of u_i
  - (1) is the relevancy condition, (2) is the exogeneity condition
  - Challenge: Exogeneity is difficult to argue (usually argue that Z_i affects Y_i only through X_i)
- This is estimated with two stage least squares (2SLS)
  - Original Equation: $Y_i = \alpha + \beta X_i + u_i$
  - First stage: $X_i = \delta_0 + \delta_1 Z_i + v_i \rightarrow$ Get $\widehat{X}_i = \hat{\delta}_0 + \hat{\delta}_1 Z_i$ ($\widehat{X}_i$ is uncorrelated with u_i)
  - Second stage: $Y_i = \beta_0 + \beta_1 \widehat{X}_i + u_i$

Instrumental variables using natural environment

- Source: Brückner, Ciccone, "Rain and the Democratic Window of Opportunity", *Econometrica*, 79(3), 923-947

- Question: Can transitory shocks to income (X_i) lead to democratic changes (Y_i)?

- Income shocks can be endogenous: Places with higher income are likely more democratic to begin with
  - Use (lack of) rainfall (Z_i): Correlates with income through agricultural productivity but unlikely to affect democratic transitions alone
  - Plausible: Very hard for rainfall itself to lead to democratization

- Result: 1% transitory negative income shock $\rightarrow$ 1.3 percentage point rise in democratic transitions

Instrumental variables using natural environment

	Democratic Transition			Democratization Step		
	(1) LS	(2) 2SLS	(3) 2SLS	(4) LS	(5) 2SLS	(6) 2SLS
Log GDP, $t-1$	0.056	−1.285**		−0.053	−1.471**	
	(0.058)	[0.027]		(0.051)	[0.029]	
Country specific recession, $t-1$			0.235**			0.279**
			[0.027]			[0.029]
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Country time trend	Yes	Yes	Yes	Yes	Yes	Yes
Common time effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	700	700	700	867	867	867
First Stage for Log GDP per capita/Country Specific Recession, $t-1$						
Log rainfall, $t-1$		0.095***	−0.519***		0.094***	−0.494***
		(0.037)	(0.164)		(0.032)	(0.151)
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Country time trend	Yes	Yes	Yes	Yes	Yes	Yes
Common time effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	700	700	700	867	867	867

Natural experiments: Difference-in-differences

- Assume two time periods (before and after) and two groups (treated and controlled)

- You have the following regression

$$Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 P_{it} + \beta_3 D_{it} P_{it} + u_{it}$$

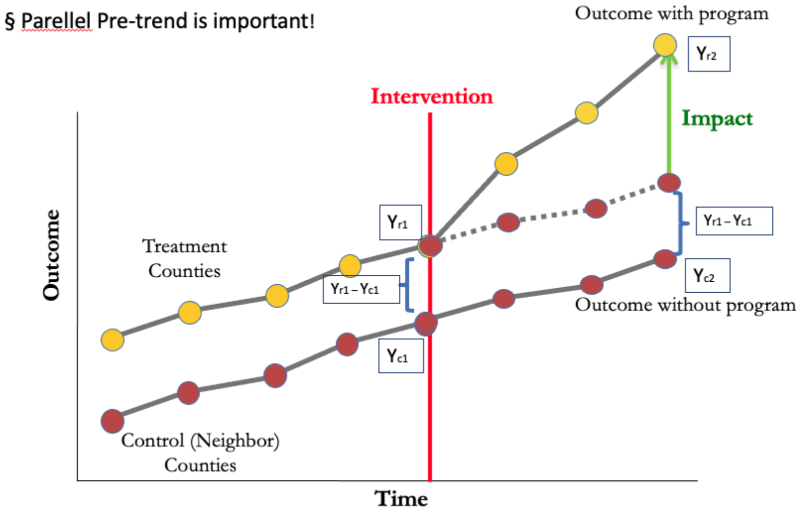
  where $D = 1$ if in treatment group, $P = 1$ if after treatment started

- You can get the following group averages
  - Treated, After: $E[Y_{it}|D_{it} = 1, P_{it} = 1] = \beta_0 + \beta_1 + \beta_2 + \beta_3$
  - Treated, Before: $E[Y_{it}|D_{it} = 1, P_{it} = 0] = \beta_0 + \beta_1$
  - Controlled, After: $E[Y_{it}|D_{it} = 0, P_{it} = 1] = \beta_0 + \beta_2$
  - Controlled, Before: $E[Y_{it}|D_{it} = 0, P_{it} = 1] = \beta_0$

- DID: $[\bar{Y}_{\text{treat,after}} - \bar{Y}_{\text{treat,before}}] - [\bar{Y}_{\text{control,after}} - \bar{Y}_{\text{control,before}}] \to \hat{\beta}_3$

- Can be generalized to many time periods and many groups with TWFE, Event-studies etc (active area of research)

Natural experiments: Conceptualizing Difference-in-differences



Natural experiments: Difference-in-Differences

- Source: Andreas V. Jensen (2025) "Educating for Democracy? Going to College Increases Political Participation", *British Journal of Political Science*, 55,1-12

- Question: Does going to college increase political participation?

- Method: Uses a survey of US students who graduated high school at 2004, observe differences in election participation in 2004 and 2008

$$\text{Turnout}_{it} = \alpha_i + \delta_t + \beta[\text{College}_i \times \text{Post}_t] + \gamma X_{it} + u_{it}$$

  - Here α_i and δ_t are individual and time fixed effects
    (Can you find their equivalent variables in the previous equation?)
  - Treated: Those who entered college between 04-08
  - Control: Did not go to college

- Result: College enrollment increase particpation by 13 percentage points

Natural experiments: Difference-in-Differences

Table 1. Effect of College Education on Voter Turnout

	Full Sample		
	(1)	(2)	(3)
Attended College × Post Period	0.137*** (0.014)	0.130*** (0.016)	0.107*** (0.016)
Time FEs & Individual FEs	✓	✓	✓
Pre-college Voting × Time FEs	✓	✓	✓
Time-varying Controls		✓	✓
Additional Pre-college Covariates × Time FEs			✓
Units	10,426	9,652	9,554
Observations	19,813	18,298	18,118

Natural experiments: Regression discontinuity

- Comparing individuals just entering the treatment vs marginally did not receive one
  - Assume you are treated if $Z_i \geq c$ and not if otherwise
  - Then, you run this regression on $Z_i \in [c - h, c + h]$ for some small h

$$Y_i = \beta X_i + \gamma(1[Z_i \geq c] \times X_i) + \delta(X_i \times (Z_i - c)) + \mu(X_i \times (Z_i - c) \times 1[Z_i \geq c]) + u_i$$

    For simplicity, assume that no other independent variables are in place: $X_i = 1$
  - $1[Z_i \geq c] = 1$ if i passes threshold for eligibility (0 if otherwise)
  - γ captures the treatment effect: Difference between those in and out of treatment
  - We also allow slope to be different under and above the cutoff: δ for people below cutoff and $\delta + \mu$ for those above

- Treatment effect estimated here is local: May not hold for out-sample

- There may be cases where threshold is not strict (fuzzy RD)

Regression discontinuity in use with electoral rules

- Source: Thomas Fujiwara (2011) “A Regression Discontinuity Test of Strategic Voting and Duverger’s Law”, *Quarterly Journal of Political Science*, 6, 197-233
- Question: Do third-party candidates perform well in majoritarian elections with run-offs than ones with no run-offs?
- Design: In Brazil, precincts with more than 200,000 registered voters have run-off elections for mayors (otherwise, no run-offs)
- Findings: Third-party candidates perform signficantly worse in places without run-offs, suggesting that voters consider winning probabilities of candidates into account

Natural experiments: Regression discontinuity